UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

SCHOOL OF SCIENCE AND TECHNOLOGY

BRAZILIAN E-COMMERCE BUSINESS INTELLIGENCE DASHBOARD

DSA 3050A – BUSINESS INTELLIGENCE AND VISUALIZATION

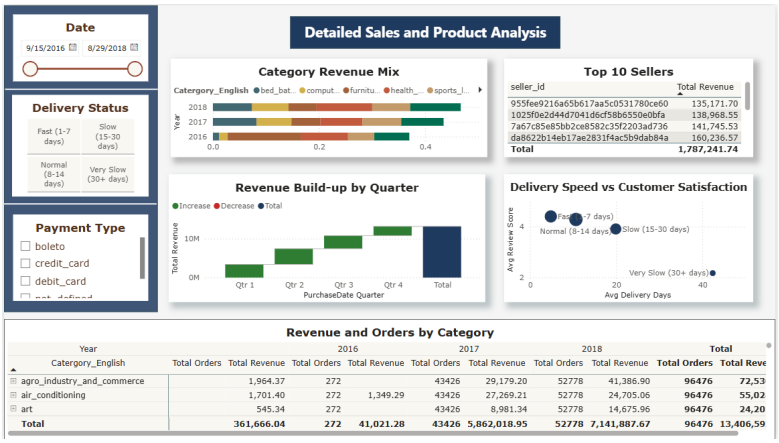
FAITH MWANGI

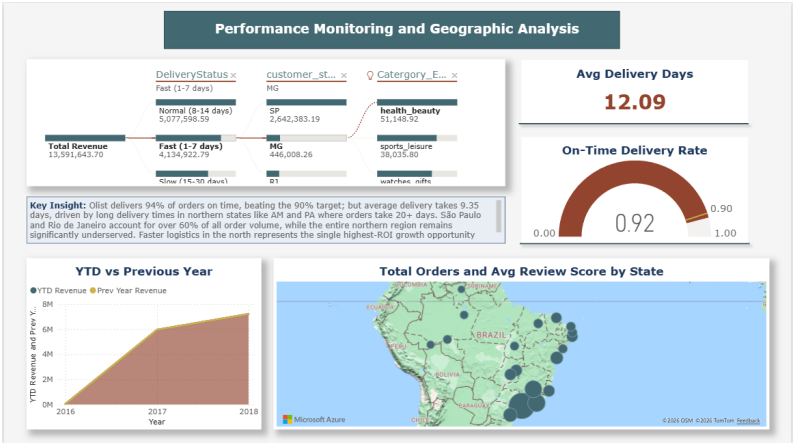
ID: -954

APRIL 2026

Executive Summary

Analysis Dashboards:





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Key Findings:

- Total revenue: R$13.59M across 96K orders (2016–2018)
- Average review score: 4.09/5, strong customer satisfaction

- On-time delivery rate: 94%, exceeding 90% target
- Health beauty and watches gifts are top revenue categories
- São Paulo dominates order volume at 40%+ of total orders

1. Introduction

1.1 Background

Olist is a Brazilian e-commerce marketplace that connects small and medium-sized businesses to customers across Brazil through a single unified platform. The dataset used in this project covers orders placed between 2016 and 2018, spanning over 99,000 transactions across 27 Brazilian states and covering product categories ranging from health and beauty to electronics, furniture, and sporting goods.

The dataset is publicly available on Kaggle and was provided by Olist under the CC BY-NC-SA 4.0 licence. It consists of eight related CSV files covering orders, order items, payments, reviews, products, customers, geolocation, and product category translations.

1.2 Problem Statement

The purpose of this project is to design and build a comprehensive Power BI business intelligence solution using the Olist e-commerce dataset. The dashboard consolidates eight data sources into a single interactive analytical platform that enables stakeholders to monitor revenue performance, evaluate delivery efficiency, assess customer satisfaction, analyse product category trends, and understand geographic demand distribution across Brazil.

1.3 Intended Audience

This dashboard has been designed to serve three primary stakeholder groups:

Executive management is the primary audience for Page 1, the Executive Summary dashboard. This audience requires high-level KPI visibility, revenue trends, and category performance summaries to support strategic planning and investment decisions.

The logistics and operations team is the primary audience for Page 3, the Performance Monitoring and Geographic Analysis dashboard. This team requires operational metrics such

as average delivery days, on-time delivery rates, and geographic demand distribution to identify bottlenecks and improve last-mile delivery performance.

The product and category management team is the primary audience for Page 2, the Detailed Sales and Product Analysis dashboard. This team requires granular visibility into category-level revenue performance, seller contributions, year-on-year category mix shifts, and the relationship between delivery performance and customer satisfaction scores.

1.4 Questions This Dashboard Answers

This Power BI solution was designed to answer the following key business questions:

1. Which product categories generate the most revenue, and is the current revenue concentration a strategic risk?
2. How does delivery performance directly affect customer review scores and overall satisfaction?
3. Which payment methods do customers prefer, and what are the implications for cash flow management?
4. Which sellers are driving the most revenue, and how is seller performance distributed across the platform?
5. Is the 90% on-time delivery target being met, and how has performance trended over time?

1.5 Structure of This Report

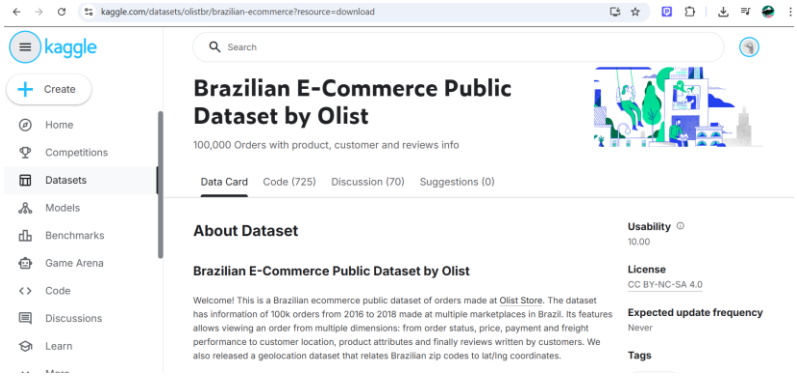
- Section 2 presents the dataset description including all source files
- Section 3 documents Power Query transformation steps applied to the raw data, including the reasoning behind each transformation and supporting evidence.
- Section 4 describes the star schema data model, including table roles, relationship cardinality, cross-filter directions
- Section 5 presents all DAX measures calculated columns, including formulas, explanations, and decision-making applications
- Section 6 provides full screenshots and detailed descriptions of all three dashboard pages

- Section 7 presents analytical insights derived from the dashboard and actionable business recommendations supported by the data.
- Section 8 provides the conclusion summarising

2. Dataset Description

2.1 Source and Licence

The dataset used in this project is the Brazilian E-Commerce Public Dataset published by Olist and made publicly available on Kaggle.



URL: kaggle.com/datasets/olistbr/brazilian-ecommerce

License: CC BY-NC-SA 4.0

2.2 Dataset Files

This dataset contains multiple related tables linked by shared keys including order_id, customer_id, and product_id, enabling the construction of a proper star schema with clearly defined fact and dimension tables.

The main orders table contains 99,441 rows.

olist_orders_dataset.csv

1	order_id	customer_id	order_status	order_purchase_timestamp	order_approved_a	order_delivered_c	order_delivered_customer_d	order_estimated_delivery_date
2	e481f51cbdc54678b7cc49136	9ef432eb6251297304e76186b10a92	delivered	02/10/2017 10:56	02/10/2017 11:07	04/10/2017 19:55	10/10/2017 21:25	18/10/2017 00:00
3	53cdb2fc8bc7dce0b6741e215	b0830fb4747a6c6d20dea0b8c802d	delivered	24/07/2018 20:41	26/07/2018 03:24	26/07/2018 14:31	07/08/2018 15:27	13/08/2018 00:00
4	47770eb9100c2d0c44946d9cf	41ce2a54c0b03bf3443c3d931a367	delivered	08/08/2018 08:38	08/08/2018 08:55	08/08/2018 13:50	17/08/2018 18:06	04/09/2018 00:00
5	949d5b44dbf5de918fe9c16f97	f88197465ea7920adcdbec7375364	delivered	18/11/2017 19:28	18/11/2017 19:45	22/11/2017 13:39	02/12/2017 00:28	15/12/2017 00:00
6	ad21c59c0840e6cb83a9ceb5	8ab97904e6daea8866dbdbc4fb7a	delivered	13/02/2018 21:18	13/02/2018 22:20	14/02/2018 19:46	16/02/2018 18:17	26/02/2018 00:00

olist_order_items_dataset.csv

1	order_id	order_iten	product_id	seller_id	shipping_limit_date	price	freight_value
2	00010242fe8c5a6d1ba	1	4244733e06e7ecb497	48436dade18ac8b2b	19/09/2017 09:45	58.9	13.29
3	00018f77f2f0320c5571	1	e5f2d52b802189ee65	dd7ddc04e1b6c2c61	03/05/2017 11:05	239.9	19.93
4	000229ec398224ef6ca	1	c777355d18b72b67a	5b51032eddd242adc	18/01/2018 14:48	199	17.87
5	00024acbcdf0a6daa1	1	7634da152a4610f159	9d7a1d34a50524090	15/08/2018 10:10	12.99	12.79
6	00042b26cf59d7ce69c	1	ac6c3623068f30de03	df560393f3a51e7455	13/02/2017 13:57	199.9	18.14

olist_order_payments_dataset.csv

	A	B	C	D	E	F
1	order_id	payment_s	payment_type	payment_i	payment_value	
2	b81ef226f3fe1789b1e8b2acac83	1	credit_card	8	99.33	
3	a9810da82917af2d9aefd1278f1(	1	credit_card	1	24.39	
4	25e8ea4e93396b6fa0d3dd708e7	1	credit_card	1	65.71	
5	ba78997921bbcdc1373bb41e91:	1	credit_card	8	107.78	
6	42fdf880ba16b47b59251dd489d	1	credit_card	2	128.45	

olist_order_reviews_dataset.csv

	A	B	C	D	E	F	G
1	review_id	order_id	review_sc	review_comment_titl	review_comment_message	review_creation_date	review_answer_timestamp
2	7bc2406110b926393a	73fc7af87114b39712e	4			18/01/2018 00:00	18/01/2018 21:46
3	80e641a11e56f04c1ad	a548910a1c6147796b	5			10/03/2018 00:00	11/03/2018 03:05
4	228ce5500dc1d8e020	f9e4b658b201a9f2ecd	5			17/02/2018 00:00	18/02/2018 14:36
5	e64fb393e7b32834bb7	658677c97b385a9be1	5		Recebi bem antes do prazo estipulado.	21/04/2017 00:00	21/04/2017 22:02
6	f7c4243c7fe1938f181b	8e6bfb81e283fa7e4f11	5		ParabÃ©ns lojas lannister adorei compr	01/03/2018 00:00	02/03/2018 10:26

olist_products_dataset.csv

	A	B	C	D	E	F	G	H	I	J
1	product_id	product_category_name	product_n	product_d	product_p	product_w	product_le	product_h	product_width_cm	
2	1e9e8ef04dbcff4541ed26657ea5	perfumaria	40	287	1	225	16	10	14	
3	3aa071139cb16b67ca9e5dea64	artes	44	276	1	1000	30	18	20	
4	96bd76ec8810374ed1b65e2919	esporte_lazer	46	250	1	154	18	9	15	
5	cef67bcfe19066a932b7673e239	bebes	27	261	1	371	26	4	26	
6	9dc1a7de274444849c219cff195	utilidades_domesticas	37	402	4	625	20	17	13	

olist_customers_dataset.csv

	A	B	C	D	E	F
1	customer_id	customer_unique_id	customer_zip_code_prefix	customer_city	customer_state	
2	06b8999e2fba1a1fbc88172c00b	861eff4711a542e4b93843c6dd7f	14409	franca	SP	
3	18955e83d337fd6b2def6b18a42	290c77bc529b7ac935b93aa66c3	09790	sao bernardo do campo	SP	
4	4e7b3e00288586ebd08712fdd03	060e732b5b29e8181a18229c7b0	01151	sao paulo	SP	
5	b2b6027bc5c5109e529d4dc635	259dac757896d24d7702b9acbb	08775	mogi das cruzes	SP	
6	4f2d8ab171c80ec8364f7c12e35	345ecd01c38d18a9036ed96c73	13056	campinas	SP	

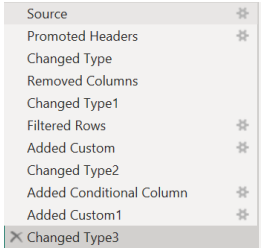
product_category_name_translation.csv

	A	B	C
1	product_category_name	product_category_name_english	
2	beleza_saude	health_beauty	
3	informatica_acessorios	computers_accessories	
4	automotivo	auto	
5	cama_mesa_banho	bed_bath_table	
6	moveis_decoracao	furniture_decor	

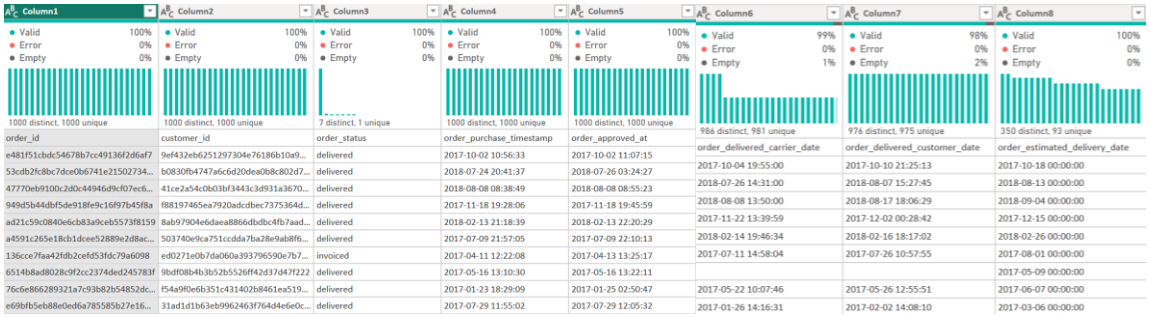
3. Data Cleaning and Transformation

3.1 FactOrders

Steps Pane:

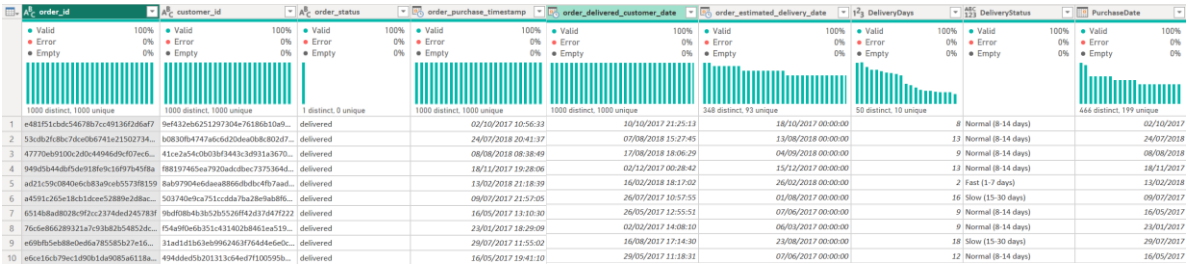


Before:



Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 99%	Valid 98%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 1%	Empty 2%	Empty 0%
1000 distinct, 1000 unique	1000 distinct, 1000 unique	7 distinct, 1 unique	1000 distinct, 1000 unique	1000 distinct, 1000 unique	986 distinct, 981 unique	976 distinct, 975 unique	350 distinct, 93 unique
order_id	customer_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date	order_delivered_customer_date	order_estimated_delivery_date
e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a9...	delivered	2017-10-02 10:56:33	2017-10-02 11:07:15	2017-10-04 19:55:00	2017-10-10 21:25:13	2017-10-18 00:00:00
53cdb2fc8bc7dce0b6741e21502734...	b0830fb4747a6c6d20dea0b8c802d7...	delivered	2018-07-24 20:41:37	2018-07-26 03:24:27	2018-07-26 14:31:00	2018-08-07 15:27:45	2018-08-13 00:00:00
47770eb9100c2d0c44946d9cf07ec6...	41ce2a54c0b03bf3443c3d931a3670...	delivered	2018-08-08 08:38:49	2018-08-08 08:55:23	2018-08-08 13:50:00	2018-08-17 18:06:29	2018-09-04 00:00:00
949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d...	delivered	2017-11-18 19:28:06	2017-11-18 19:45:59	2017-11-22 13:39:59	2017-12-02 00:28:42	2017-12-15 00:00:00
ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad...	delivered	2018-02-13 21:18:39	2018-02-13 22:20:29	2018-02-14 19:46:34	2018-02-16 18:17:02	2018-02-26 00:00:00
a4591c265e18cb1dcee52889e2d8ac...	503740e9ca751ccdda7ba28e9ab8f6...	delivered	2017-07-09 21:57:05	2017-07-09 22:10:13	2017-07-11 14:58:04	2017-07-26 10:57:55	2017-08-01 00:00:00
136cce7faa42fdb2cefd53fdc79a6098	ed0271e0b7da060a393796590e7b7...	invoiced	2017-04-11 12:22:08	2017-04-13 13:25:17			2017-05-09 00:00:00
6514b8ad8028c9f2cc2374ded245783f	9bdf08b4b3b52b5526ff42d37d47f222	delivered	2017-05-16 13:10:30	2017-05-16 13:22:11	2017-05-22 10:07:46	2017-05-26 12:55:51	2017-06-07 00:00:00
76c6e866289321a7c93b82b54852dc...	f54a9f0e6b351c431402b8461ea519...	delivered	2017-01-23 18:29:09	2017-01-25 02:50:47	2017-01-26 14:16:31	2017-02-02 14:08:10	2017-03-06 00:00:00
e69bfb5eb88e0ed6a785585b27e16...	31ad1d1b63eb9962463f764d4e6e0c...	delivered	2017-07-29 11:55:02	2017-07-29 12:05:32			

After:



	order_id	customer_id	order_status	order_purchase_timestamp	order_delivered_customer_date	order_estimated_delivery_date	DeliveryDays	DeliveryStatus	PurchaseDate
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
	1000 distinct, 1000 unique	1000 distinct, 1000 unique	1 distinct, 0 unique	1000 distinct, 1000 unique	1000 distinct, 1000 unique	348 distinct, 93 unique	50 distinct, 10 unique		466 distinct, 199 unique
1	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a9...	delivered	02/10/2017 10:56:33	10/10/2017 21:25:13	18/10/2017 00:00:00	8	Normal (8-14 days)	02/10/2017
2	53cdb2fc8bc7dce0b6741e21502734...	b0830fb4747a6c6d20dea0b8c802d7...	delivered	24/07/2018 20:41:37	07/08/2018 15:27:45	13/08/2018 00:00:00	13	Normal (8-14 days)	24/07/2018
3	47770eb9100c2d0c44946d9cf07ec6...	41ce2a54c0b03bf3443c3d931a3670...	delivered	08/08/2018 08:38:49	17/08/2018 18:06:29	04/09/2018 00:00:00	9	Normal (8-14 days)	08/08/2018
4	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d...	delivered	18/11/2017 19:28:06	02/12/2017 00:28:42	15/12/2017 00:00:00	13	Normal (8-14 days)	18/11/2017
5	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad...	delivered	13/02/2018 21:18:39	16/02/2018 18:17:02	26/02/2018 00:00:00	2	Fast (1-7 days)	13/02/2018
6	a4591c265e18cb1dcee52889e2d8ac...	503740e9ca751ccdda7ba28e9ab8f6...	delivered	09/07/2017 21:57:05	26/07/2017 10:57:55	01/08/2017 00:00:00	16	Slow (15-30 days)	09/07/2017
7	6514b8ad8028c9f2cc2374ded245783f	9bdf08b4b3b52b5526ff42d37d47f222	delivered	16/05/2017 13:10:30	26/05/2017 12:55:51	07/06/2017 00:00:00	9	Normal (8-14 days)	16/05/2017
8	76c6e866289321a7c93b82b54852dc...	f54a9f0e6b351c431402b8461ea519...	delivered	23/01/2017 18:29:09	02/02/2017 14:08:10	06/03/2017 00:00:00	9	Normal (8-14 days)	23/01/2017
9	e69bfb5eb88e0ed6a785585b27e16...	31ad1d1b63eb9962463f764d4e6e0c...	delivered	29/07/2017 11:55:02	16/08/2017 17:14:30	23/08/2017 00:00:00	18	Slow (15-30 days)	29/07/2017
10	e6ce16cb79ec1d90b1da9085a6118a...	494dded5b201313c64ed7f100595b...	delivered	16/05/2017 19:41:10	29/05/2017 11:18:31	07/06/2017 00:00:00	12	Normal (8-14 days)	16/05/2017

1. **Remove unused columns from Orders:**

   These intermediate logistics timestamps are not needed, removing them reduces model size and simplifies the schema.

   order_approved_at and order_delivered_carrier_date

```
= Table.RemoveColumns(#"Changed Type",{"order_approved_at", "order_delivered_carrier_date"})
```

2. **Change data types**

```
= Table.TransformColumnTypes(#"Removed Columns",{{"order_id", type text}, {"customer_id", type text},
    {"order_status", type text}, {"order_purchase_timestamp", type datetime}, {"order_delivered_customer_date",
    type datetime}, {"order_estimated_delivery_date", type datetime}})
```

3. **Remove rows with null delivery date**

A small number of "delivered" rows still have a null delivery date, data entry errors. These prevent DeliveryDays from calculating.

```
= Table.SelectRows(#"Changed Type1", each ([order_delivered_customer_date] <> null))
```

4. **Create DeliveryDays custom column**

Core metric for delivery performance analysis. Converts two timestamps into a single interpretable integer.

```
= Table.AddColumn(#"Filtered Rows", "DeliveryDays", each Duration.Days([order_delivered_customer_date] -
    [order_purchase_timestamp]))
```

5. **Create DeliveryStatus conditional column**

Converts continuous number to categorical label. Enables slicers, scatter chart legends, and percentage-of-category DAX filtering.

```
= Table.AddColumn(#"Changed Type2", "DeliveryStatus", each if [DeliveryDays] <= 7 then "Fast (1-7 days)" else if
    [DeliveryDays] <= 14 then "Normal (8-14 days)" else if [DeliveryDays] <= 30 then "Slow (15-30 days)" else "Very
    Slow (30+ days)")
```

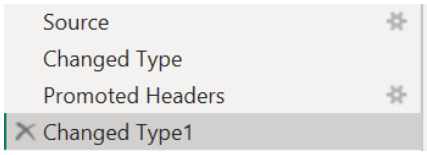
6. **Extract PurchaseDate from timestamp**

DimDate contains Date values with no time component. The relationship requires both sides to be the same type, Date.From strips the time part.

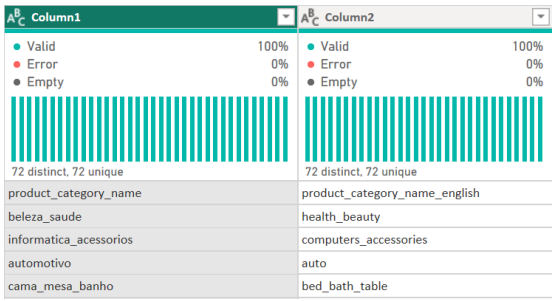
```
= Table.AddColumn(#"Added Conditional Column", "PurchaseDate", each Date.From([order_purchase_timestamp]))
```

3.2 CategoryTranslation

Steps Pane:

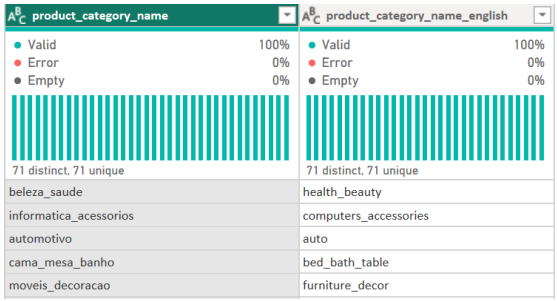


Before:



Column1	Column2
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
72 distinct, 72 unique	72 distinct, 72 unique
product_category_name	product_category_name_english
beleza_saude	health_beauty
informatica_acessorios	computers_accessories
automotivo	auto
cama_mesa_banho	bed_bath_table

After:



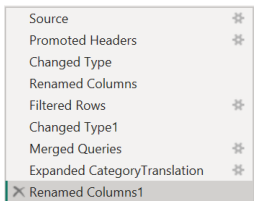
product_category_name	product_category_name_english
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
71 distinct, 71 unique	71 distinct, 71 unique
beleza_saude	health_beauty
informatica_acessorios	computers_accessories
automotivo	auto
cama_mesa_banho	bed_bath_table
moveis_decoracao	furniture_decor

1. **Use First Row as Headers**

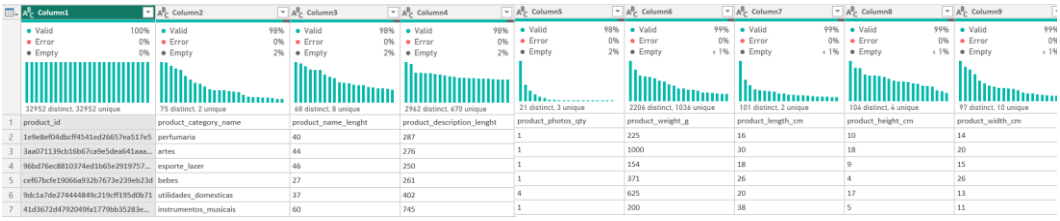
```
= Table.TransformColumnTypes(#"Promoted Headers",{{"product_category_name", type text},
    {"product_category_name_english", type text}})
```

3.3 DimProducts

Steps Pane:

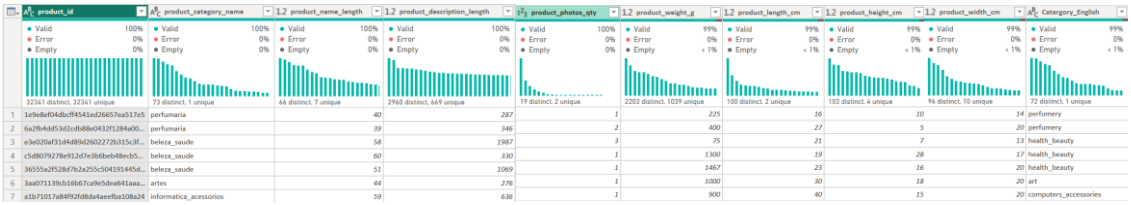


Before:



	Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9
	Valid 100%	Valid 98%	Valid 98%	Valid 98%	Valid 98%	Valid 99%	Valid 99%	Valid 99%	Valid 99%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 2%	Empty 2%	Empty 2%	Empty 2%	Empty < 1%	Empty < 1%	Empty < 1%	Empty < 1%
	32952 distinct, 32952 unique	75 distinct, 2 unique	68 distinct, 8 unique	2962 distinct, 670 unique	21 distinct, 3 unique	2206 distinct, 1036 unique	101 distinct, 2 unique	104 distinct, 4 unique	97 distinct, 10 unique
1	product_id	product_category_name	product_name_lenght	product_description_lenght	product_photos_qty	product_weight_g	product_length_cm	product_height_cm	product_width_cm
2	1e9e8ef04dbcff4541ed26657ea517e5	perfumaria	40	287	1	225	16	10	14
3	3aa071139cb16b67ca9e5dea641aaa...	artes	44	276	1	1000	30	18	20
4	96bd76ec8810374ed1b65e2919757...	esporte_lazer	46	250	1	154	18	9	15
5	cef67bcfe19066a932b7673e239eb23d	bebes	27	261	1	371	26	4	26
6	9dc1a7de274444849c219cff195d0b71	utilidades_domesticas	37	402	4	625	20	17	13
7	41d3672d4792049fa1779bb35283e...	instrumentos_musicais	60	745	1	200	38	5	11

After:



	product_id	product_category_name	product_name_length	product_description_length	product_photos_qty	product_weight_g	product_length_cm	product_height_cm	product_width_cm	Catergory_English
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 99%	Valid 99%	Valid 99%	Valid 99%	Valid 99%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty < 1%	Empty < 1%	Empty < 1%	Empty < 1%	Empty < 1%
	32341 distinct, 32341 unique	73 distinct, 1 unique	66 distinct, 7 unique	2960 distinct, 669 unique	19 distinct, 2 unique	2203 distinct, 1039 unique	100 distinct, 2 unique	103 distinct, 4 unique	96 distinct, 10 unique	72 distinct, 1 unique
1	1e9e8ef04dbcff4541ed26657ea517e5	perfumaria	40	287	1	225	16	10	14	perfumery
2	6a2fb4dd53d2cdb88e0432f1284a00...	perfumaria	39	346	2	400	27	5	20	perfumery
3	e3e020af31d4d89d2602272b315c3f...	beleza_saude	58	1987	3	75	21	7	13	health_beauty
4	c5d8079278e912d7e3b6beb48ecb5...	beleza_saude	60	330	1	1300	19	28	17	health_beauty
5	36555a2f528d7b2a255c50419145d...	beleza_saude	51	1069	1	1467	23	16	20	health_beauty
6	3aa071139cb16b67ca9e5dea641aaa...	artes	44	276	1	1000	30	18	20	art
7	a1b71017a84f92fd8da4aeefba108a24	informatica_acessorios	59	636	1	900	40	15	20	computers_accessories

1. **Remove Empty Rows**

```
= Table.SelectRows(#"Renamed Columns", each [product_category_name] <> null and [product_category_name] <> "")
```

2. **Change Data Types**

```
= Table.TransformColumnTypes(#"Filtered Rows",{{"product_height_cm", type number}, {"product_width_cm", type number},
    {"product_length_cm", type number}, {"product_weight_g", type number}, {"product_description_length", type number},
    {"product_name_length", type number}})
```

3. **Merge CategoryTranslation into Products**

   All category names are in Portuguese. English names are required for readable dashboard visuals. Null replacement prevents blank chart labels.

   a. **Merge tables**

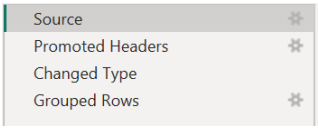
```
= Table.NestedJoin(#"Changed Type1", {"product_category_name"}, CategoryTranslation, {"product_category_name"},
    "CategoryTranslation", JoinKind.LeftOuter)
```

   b. **Expand Merged Columns**

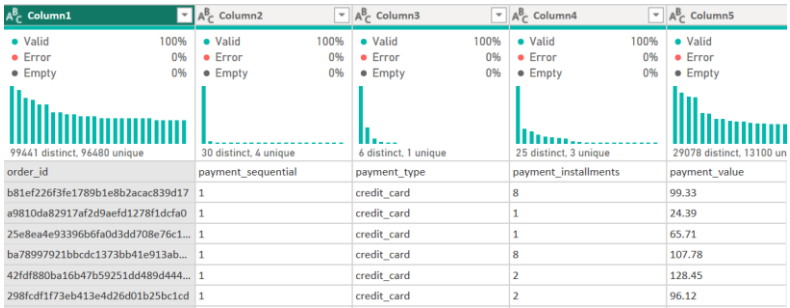
```
= Table.ExpandTableColumn(#"Merged Queries", "CategoryTranslation", {"product_category_name_english"},
    {"product_category_name_english"})
```

3.4 DimOrderPayments

Steps Pane:

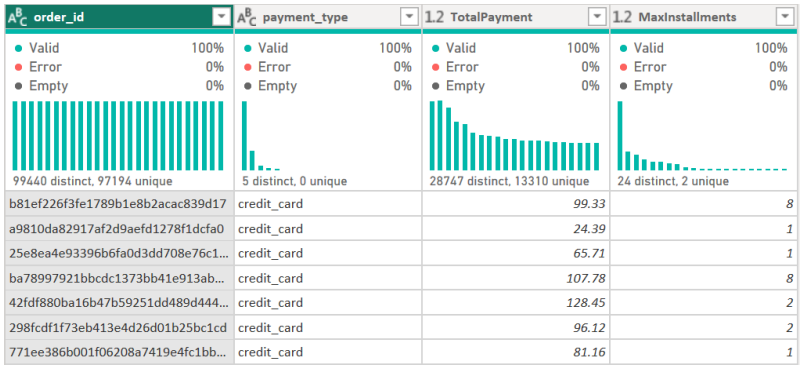


Before:



Column1	Column2	Column3	Column4	Column5
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid
Error 0%	Error 0%	Error 0%	Error 0%	Error
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty
99441 distinct, 96480 unique	30 distinct, 4 unique	6 distinct, 1 unique	25 distinct, 3 unique	29078 distinct, 13100 un
order_id	payment_sequential	payment_type	payment_installments	payment_value
b81ef226f3fe1789b1e8b2acac839d17	1	credit_card	8	99.33
a9810da82917af2d9aefd1278f1dcfa0	1	credit_card	1	24.39
25e8ea4e93396b6fa0d3dd708e76c1...	1	credit_card	1	65.71
ba78997921bbcdc1373bb41e913ab...	1	credit_card	8	107.78
42fdf880ba16b47b59251dd489d444...	1	credit_card	2	128.45
298fcdf1f73eb413e4d26d01b25bc1cd	1	credit_card	2	96.12

After:



order_id	payment_type	TotalPayment	MaxInstallments
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
99440 distinct, 97194 unique	5 distinct, 0 unique	28747 distinct, 13310 unique	24 distinct, 2 unique
b81ef226f3fe1789b1e8b2acac839d17	credit_card	99.33	8
a9810da82917af2d9aefd1278f1dcfa0	credit_card	24.39	1
25e8ea4e93396b6fa0d3dd708e76c1...	credit_card	65.71	1
ba78997921bbcdc1373bb41e913ab...	credit_card	107.78	8
42fdf880ba16b47b59251dd489d444...	credit_card	128.45	2
298fcdf1f73eb413e4d26d01b25bc1cd	credit_card	96.12	2
771ee386b001f06208a7419e4fc1bb...	credit_card	81.16	1

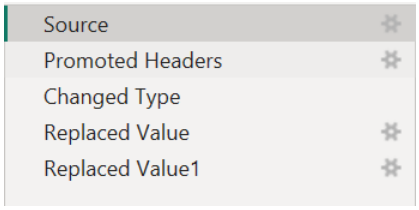
1. **Aggregate Payments to one row per order**

   The Payments table has multiple rows per order_id. Without aggregation the relationship to Orders would be many-to-many, corrupting all payment measures.

```
= Table.Group(#"Changed Type", {"order_id", "payment_type"}, {{"TotalPayment", each List.Sum([payment_value]), type
    nullable number}, {"MaxInstallments", each List.Max([payment_installments]), type nullable number}})
```

3.5 DimOrderReviews

Steps Pane:



Before:

Column1	Column2	Column3	Column4	Column5	Column6	Column7
Valid 100%	Valid 100%	Valid 100%	Valid 12%	Valid 44%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 88%	Empty 56%	Empty 0%	Empty 0%
1000 distinct, 1000 unique	1000 distinct, 1000 unique	6 distinct, 1 unique	97 distinct, 85 unique	435 distinct, 431 unique	405 distinct, 148 unique	1000 distinct, 1000 unique
review_id	order_id	review_score	review_comment_title	review_comment_message	review_creation_date	review_answer_timestamp
7bc2406110b926393aa56f80a40eba...	73fc7af87114b39712e6da79b0a377...	4			2018-01-18 00:00:00	2018-01-18 21:46:59
80e641a11e56f04c1ad469d5645fdfde	a548910a1c6147796b98fdf73dbeba...	5			2018-03-10 00:00:00	2018-03-11 03:05:13
228ce5500dc1d8e020d8d1322874b...	f9e4b658b201a9f2ecdecbb34bed03...	5			2018-02-17 00:00:00	2018-02-18 14:36:24
e64fb393e7b32834bb789ff8bb3075...	658677c97b385a9be170737859d35...	5		Recebi bem antes do prazo estipulado.	2017-04-21 00:00:00	2017-04-21 22:02:06
f7c4243c7fe1938f181bec41a392bdeb	8e6bfb81e283fa7e4f11123a3fb894f1	5		Parabéns lojas lannister adorei comprar pela Internet seguro e prático...	2018-03-01 00:00:00	2018-03-02 10:26:53
15197aa66ff4d0650b5434f1b46cda19	b18dcdf73be66366873cd26c5724d1...	1			2018-04-13 00:00:00	2018-04-16 00:39:37

After:

review_id	order_id	review_score	review_comment_title	review_comment_message	review_creation_date	review_answer_timestamp
Valid 100%	Valid 100%	Valid 100%	Valid 12%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 88%	Empty 0%	Empty 0%	Empty 0%
1000 distinct, 1000 unique	1000 distinct, 1000 unique	5 distinct, 0 unique	96 distinct, 84 unique	434 distinct, 430 unique	404 distinct, 147 unique	1000 distinct, 1000 unique
7bc2406110b926393aa56f80a40eba...	73fc7af87114b39712e6da79b0a377...	4	null	No Comment	18/01/2018 00:00:00	18/01/2018 21:46:59
80e641a11e56f04c1ad469d5645fdfde	a548910a1c6147796b98fdf73dbeba...	5	null	No Comment	10/03/2018 00:00:00	11/03/2018 03:05:13
228ce5500dc1d8e020d8d1322874b...	f9e4b658b201a9f2ecdecbb34bed03...	5	null	No Comment	17/02/2018 00:00:00	18/02/2018 14:36:24
e64fb393e7b32834bb789ff8bb3075...	658677c97b385a9be170737859d35...	5	null	Recebi bem antes do prazo estipulado.	21/04/2017 00:00:00	21/04/2017 22:02:06
f7c4243c7fe1938f181bec41a392bdeb	8e6bfb81e283fa7e4f11123a3fb894f1	5	null	Parabéns lojas lannister adorei comprar pela Int...	01/03/2018 00:00:00	02/03/2018 10:26:53
15197aa66ff4d0650b5434f1b46cda19	b18dcdf73be66366873cd26c5724d1...	1	null	No Comment	13/04/2018 00:00:00	16/04/2018 00:39:37
07f9bee5d1b850860defd761afa7ff16	e48aa0d2dcec3a2e87348811bcfdf22b	5	null	No Comment	16/07/2017 00:00:00	18/07/2017 19:30:34
7c6400515c67679fbee952a7525281ef	c31a859e34e3adac22f376954e19b3...	5	null	No Comment	14/08/2018 00:00:00	14/08/2018 21:36:06
a3f6f7f6f433de0aefbb97da197c554c	9c214ac970e84273583ab523dfafd09b	5	null	No Comment	17/05/2017 00:00:00	18/05/2017 12:05:37
8670d52e15e00043ae7de4c01cc2fe...	b9bf720beb4ab3728760088589c621...	4	recomendo	aparelho eficiente. no site a marca do aparelho...	22/05/2018 00:00:00	23/05/2018 16:45:47

1. **Replace Empty Cells**
   a. **Review Title**

```
= Table.ReplaceValue(#"Changed Type","","No Comment",Replacer.ReplaceValue,{"review_comment_message"})
```

   b. **Review/Comment**

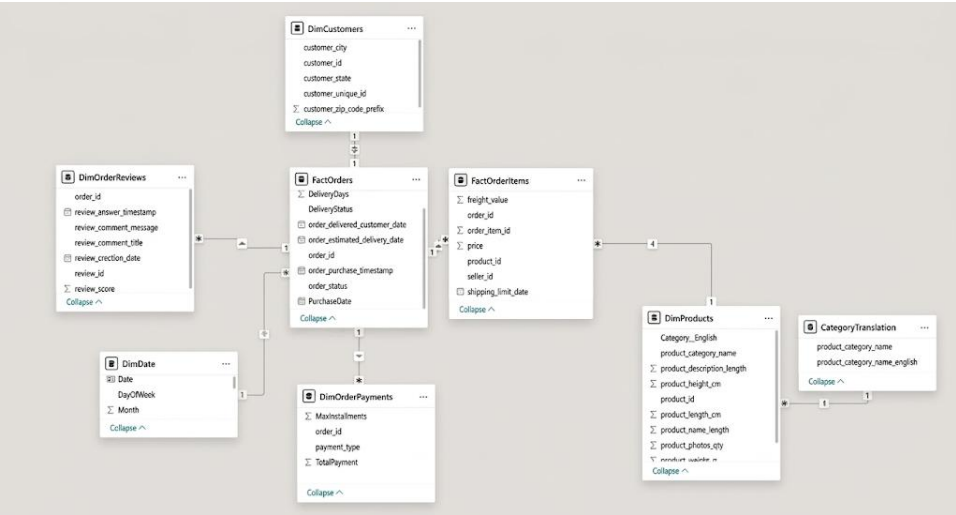
```
= Table.ReplaceValue(#"Changed Type","","No Comment",Replacer.ReplaceValue,{"review_comment_message"})
```

3.6 DimDate

PurchaseDate	Year	Month	MonthName	Quarter
Valid 100% Error 0% Empty 0% 612 distinct, 612 unique	Valid 100% Error 0% Empty 0% 3 distinct, 0 unique	Valid 100% Error 0% Empty 0% 12 distinct, 0 unique	Valid 100% Error 0% Empty 0% 12 distinct, 0 unique	Valid 100% Error 0% Empty 0% 4 distinct, 0 unique
02/10/2017	*2017*	*10*	October	*4*
24/07/2018	*2018*	*7*	July	*3*
08/08/2018	*2018*	*8*	August	*3*
18/11/2017	*2017*	*11*	November	*4*
13/02/2018	*2018*	*2*	February	*1*
09/07/2017	*2017*	*7*	July	*3*

4. Data Modeling

4.1 Overview



The model follows a star schema structure with two fact tables at the centre and five-dimension tables surrounding them. Because the model contains two fact tables sharing common dimensions it can also be described as a fact constellation schema

4.2 Fact Tables

FactOrders

This table contains the delivery metrics for the project including the DeliveryDays calculated column, the DeliveryStatus conditional column, and the PurchaseDate column used to link to the time dimension.

FactOrderItems

FactOrderItems is the revenue fact table. Each row represents one item sold within an order. This table contains the price and freight_value columns which are the source of all revenue measures in the project including Total Revenue, Total Freight Cost, and Average Order Value.

4.3 Dimension Tables

DimDate

DimDate is the time dimension. It was created in Power Query as a blank query using M language, generating one row for every calendar day between 1 September 2016 and 31 October 2018.

DimCustomers

DimCustomers is the customer and geography dimension. Each row represents one customer record

DimProducts

DimProducts is the product dimension. Each row represents one product listed on the Olist platform.

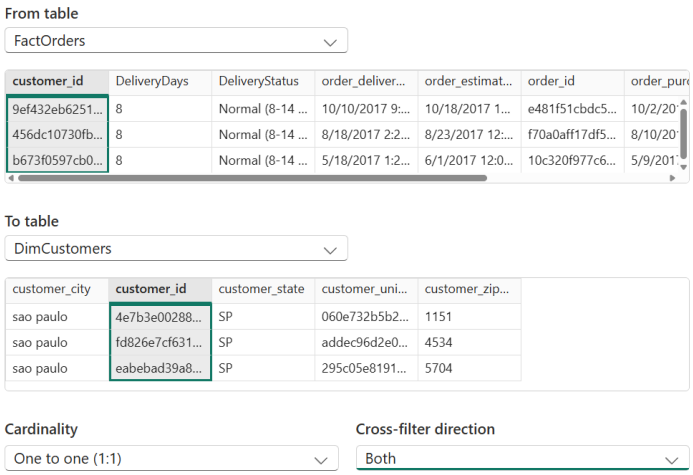
DimOrderPayments

DimOrderPayments describes how each order was paid. After aggregation in Transformation 10 each row represents one order.

DimOrderReviews

DimOrderReviews captures customer satisfaction data. Each row represents one customer review.

4.3 Relationships



FactOrders connects to DimCustomers through customer_id on both sides.

This relationship enables payment type analysis to be filtered by order-level attributes.

From table

DimOrderReviews

order_id	review_answe...	review_comm...	review_comm...	review_creati...	review_id	review_sco
46d2d651f00...	8/31/2018 3:1...	No Comment	null	8/17/2018 12:...	31b63ea961f...	5
01b4dfef1c88...	8/17/2018 10:...	No Comment	null	8/17/2018 12:...	c02f91335154...	5
9b1d0dd4286...	8/22/2018 6:3...	No Comment	null	8/17/2018 12:...	de295095fe4...	5

To table

FactOrders

customer_id	DeliveryDays	DeliveryStatus	order_deliver...	order_estimat...	order_id	order_pur
9ef432eb625...	8	Normal (8-14 ...	10/10/2017 9:...	10/18/2017 1...	e481f51cbdc5...	10/2/20
456dc10730f...	8	Normal (8-14 ...	8/18/2017 2:2...	8/23/2017 12:...	f70a0aff17df5...	8/10/20
b673f0597cb...	8	Normal (8-14 ...	5/18/2017 1:2...	6/1/2017 12:0...	10c320f977c6...	5/9/201

Cardinality

Many to one (*:1)

Cross-filter direction

Single

This relationship enables customer satisfaction scores to be analysed alongside delivery performance metrics from FactOrders.

From table

FactOrderItems

freight_value	order_id	order_item_id	price	product_id	seller_id	shipping_l
15.1	00c8be06a80...	1	19.9	5d66715cc92...	128639473a1...	2/15/20
15.1	02605457086...	1	19.9	fed5c40c27e1...	128639473a1...	12/14/2
15.1	044cd7ffb6a4...	1	19.9	62c2b9bd445...	002100f778ce...	11/7/20

To table

FactOrders

customer_id	DeliveryDays	DeliveryStatus	order_deliver...	order_estimat...	order_id	order_pur
9ef432eb625...	8	Normal (8-14 ...	10/10/2017 9:...	10/18/2017 1...	e481f51cbdc5...	10/2/20
456dc10730f...	8	Normal (8-14 ...	8/18/2017 2:2...	8/23/2017 12:...	f70a0aff17df5...	8/10/20
b673f0597cb...	8	Normal (8-14 ...	5/18/2017 1:2...	6/1/2017 12:0...	10c320f977c6...	5/9/201

Cardinality

Many to one (*:1)

Cross-filter direction

Single

This relationship allows revenue measures calculated from FactOrderItems to be filtered by any attribute on FactOrders including delivery status, purchase date, and order status.

From table

DimDate

Date	DayOfWeek	Month	MonthName	MonthShort	Quarter	WeekNo
01/07/2017	Saturday	7	July	Jul	Q3	26
02/07/2017	Sunday	7	July	Jul	Q3	27
03/07/2017	Monday	7	July	Jul	Q3	27

To table

FactOrders

Status	order_deliver...	order_estimat...	order_id	order_purcha...	order_status	PurchaseDate
(8-14 ...	10/10/2017 9:...	10/18/2017 1...	e481f51cbdc5...	10/2/2017 10:...	delivered	02/10/2017
(8-14 ...	8/18/2017 2:2...	8/23/2017 12:...	f70a0aff17df5...	8/10/2017 11:...	delivered	10/08/2017
(8-14 ...	5/18/2017 1:2...	6/1/2017 12:0...	10c320f977c6...	5/9/2017 8:48...	delivered	09/05/2017

Cardinality

One to many (1:*)

Cross-filter direction

Single

FactOrders connects to DimDate through PurchaseDate on the FactOrders side and Date on the DimDate side. This relationship enables all time-based filtering and time intelligence calculations across the model.

From table

DimOrderPayments

MaxInstallme...	order_id	payment_type	TotalPayment
1	0573b5e23cb...	boleto	51.95
1	8cd68144cdb...	boleto	330.66
1	b69b9260e79...	boleto	283.34

To table

FactOrders

customer_id	DeliveryDays	DeliveryStatus	order_deliver...	order_estimat...	order_id	order_pur
9ef432eb625...	8	Normal (8-14 ...	10/10/2017 9:...	10/18/2017 1...	e481f51cbdc5...	10/2/20
456dc10730f...	8	Normal (8-14 ...	8/18/2017 2:2...	8/23/2017 12:...	f70a0aff17df5...	8/10/20
b673f0597cb...	8	Normal (8-14 ...	5/18/2017 1:2...	6/1/2017 12:0...	10c320f977c6...	5/9/201

Cardinality

Many to one (*:1)

Cross-filter direction

Single

This relationship enables payment type analysis to be filtered by order-level attributes.

From table

FactOrderItems

freight_value	order_id	order_item_id	price	product_id	seller_id	shipping_l
15.1	00c8be06a80...	1	19.9	5d66715cc92...	128639473a1...	2/15/20
15.1	02605457086...	1	19.9	fed5c40c27e1...	128639473a1...	12/14/2
15.1	044cd7ffb6a4...	1	19.9	62c2b9bd445...	002100f778ce...	11/7/20

To table

DimProducts

Catergory_En...	product_cate...	product_desc...	product_heig...	product_id	product_lengt...	product_n
fashion_bags_...	fashion_bolsa...	770	2	5d2754abdc8...	16	53
fashion_bags_...	fashion_bolsa...	731	2	49feeab25d6...	16	48
fashion_bags_...	fashion_bolsa...	83	2	8f4735c09292...	16	60

Cardinality

Many to one (*:1)

Cross-filter direction

Single

This relationship enables all category-level revenue analysis by linking item-level prices to their corresponding product categories.

From table

DimProducts

Catergory_En...	product_cate...	product_desc...	product_heig...	product_id	product_lengt...	product_n
fashion_bags_...	fashion_bolsa...	770	2	5d2754abdc8...	16	53
fashion_bags_...	fashion_bolsa...	731	2	49feeab25d6...	16	48
fashion_bags_...	fashion_bolsa...	83	2	8f4735c09292...	16	60

To table

CategoryTranslation

product_cate...	product_cate...
beleza_saude	health_beauty
informatica_a...	computers_ac...
automotivo	auto

Cardinality

Many to one (*:1)

Cross-filter direction

Single

This relationship allows for the translation of product names to English.

5. DAX Measures and Calculated Columns

5.1 Calculated Columns

Calculated Column 1: DeliveryDays

```
Avg Delivery Days = AVERAGE(FactOrders[DeliveryDays])
```

This column calculates the number of days between the order purchase timestamp and the actual delivery date for each order, producing a whole number representing total delivery time.

Calculated Column 2: DeliveryStatus

```
= Table.AddColumn(#"Changed Type2", "DeliveryStatus", each if
    [DeliveryDays] <= 7 then "Fast (1-7 days)" else if
    [DeliveryDays] <= 14 then "Normal (8-14 days)" else if
    [DeliveryDays] <= 30 then "Slow (15-30 days)" else "Very
    Slow (30+ days)")
```

This column categorises each order's delivery time into one of four performance bands.

5.2 DAX Measures

1. **Total Revenue**

```
Total Revenue = SUM(FactOrderItems[price])
```

The sum of all item prices across all orders within the current filter context

2. **Total Freight Cost**

```
Total Freight Cost = SUM(FactOrderItems[freight_value])
```

The total shipping cost across all order items within the current filter context.

3. **Avg. Order Value**

```
Avg Order Value =
DIVIDE([Total Revenue], [Total Orders], 0)
```

The average revenue generated per order by dividing Total Revenue by Total Orders.

4. Avg. Review Score

```
Avg Review Score = AVERAGE(DimOrderReviews[review_score])
```

The mean customer review score across all reviews within the current filter context

5. Total Orders

```
Total Orders = DISTINCTCOUNT(FactOrders[order_id])
```

The mean customer review score across all reviews within the current filter context

6. YTD Revenue

```
YTD Revenue =
TOTALYTD([Total Revenue], DimDate[Date])
```

The cumulative year-to-date revenue from the first day of the current year to the last date in the current filter context.

7. Previous Year Revenue

```
Prev Year Revenue =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR(DimDate[Date])
)
```

Calculates total revenue for the same time period in the previous year using the SAMEPERIODLASTYEAR time intelligence function.

8. Revenue YoY Growth %

```
Revenue YoY Growth % =
DIVIDE(
    [Total Revenue] - [Prev Year Revenue],
    [Prev Year Revenue],
    0
)
```

This measure calculates the percentage change in revenue compared to the same period in the previous year.

9. Revenue % by Category

```
Revenue % by Category =
DIVIDE(
    [Total Revenue],
    CALCULATE([Total Revenue], ALL(DimProducts[Catergory_English])),
    0
)
```

This measure calculates each product category's share of total revenue as a percentage.

10. Category Revenue Rank

```
Category Rank =
RANKX(
    ALL(DimProducts[Catergory_English]),
    [Total Revenue],
    ,
    DESC,
    DENSE
)
```

This measure assigns a rank to each product category based on its Total Revenue, where rank 1 is the highest revenue category.

11. On-Time Delivery Rate

```
On-Time Delivery Rate =
DIVIDE(
    COUNTROWS(
        FILTER(
            FactOrders,
            FactOrders[order_delivered_customer_date]
                <= FactOrders[order_estimated_delivery_date]
        )
    ),
    [Total Orders],
    0
)
```

This measure calculates the proportion of orders that were delivered on or before the estimated delivery date.

6. Dashboard Design and Visualisation

6.1 Overview

1. Executive Dashboard

Design Choices:

Four KPI cards sit at the top showing Total Revenue, Total Orders, Average Order Value and Average Review Score.

The Top 10 Categories bar chart is sorted descending by revenue with a Top N filter of 10 applied.

The Payment Type donut chart shows the split across credit card, boleto, voucher and debit card.

The Monthly Revenue Trend line chart uses DimDate[YearMonth] on the X axis for chronological ordering.

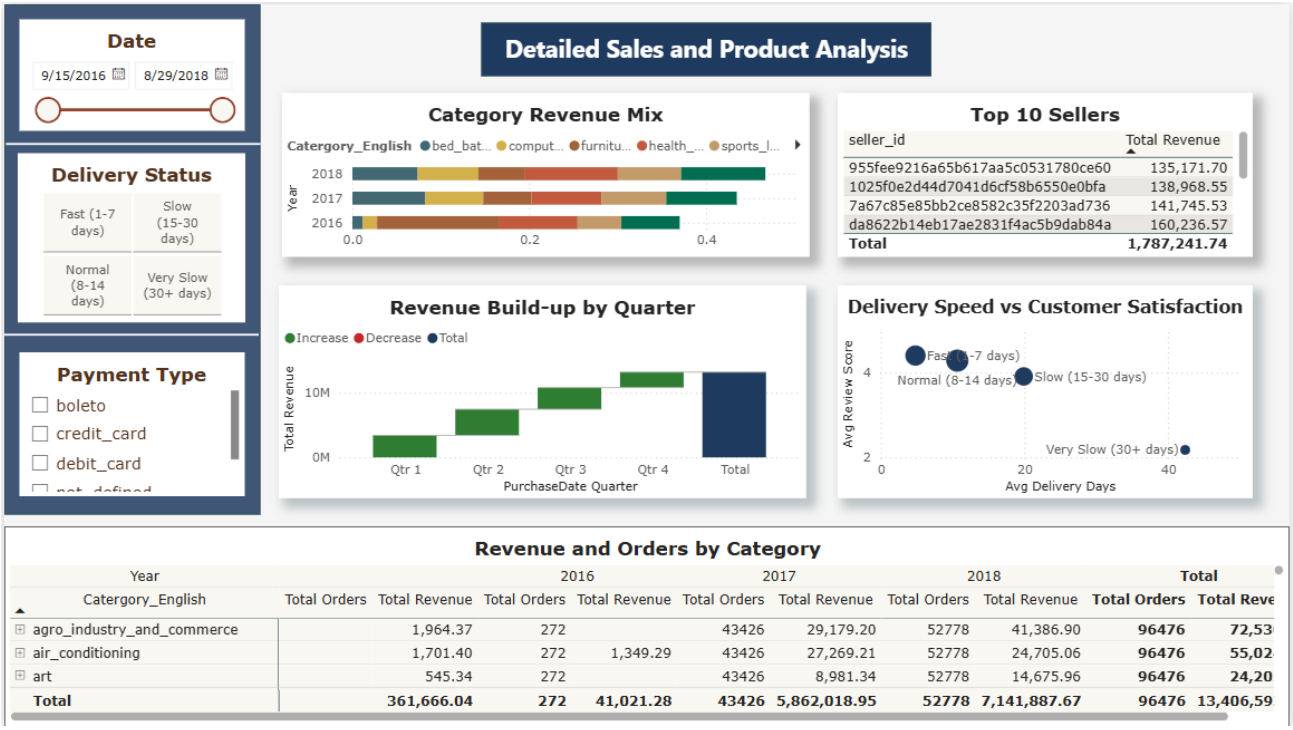
The Orders by State map uses customer_state with colour saturation encoding order volume, zoomed into Brazil.

Three slicers sit in the left panel controlling Year, Product Category and State.

How Users Interact :

Clicking a year tile, category bar, or state on the map filters all other visuals on the page. The Product Category dropdown supports text search. The Reset button returns the page to its default view.

2. Detailed Sales and Product Analysis



Design Choices

The scatter chart plots Avg Delivery Days on the X axis, Avg Review Score on the Y axis, Total Revenue as bubble size, and DeliveryStatus as the legend. Each bubble is a product category.

The matrix table has Category_English on rows with drill-down to product level, Year on columns, and data bars applied to the Total Revenue column.

The waterfall chart shows revenue accumulation by YearQuarter with green bars for increases and navy for the cumulative total.

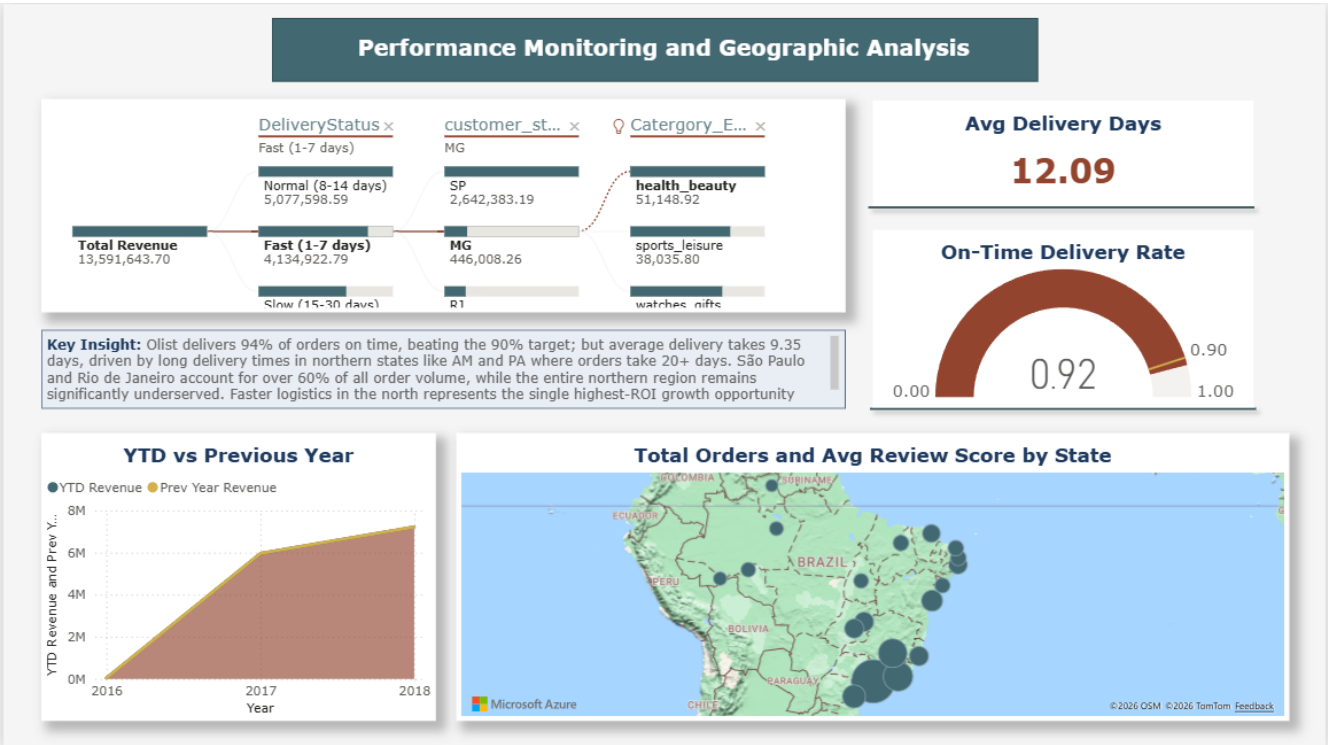
The Top 10 Sellers table uses a Top N filter of 10 ranked by Total Revenue.

The 100% stacked bar chart shows category revenue mix shifts across 2016, 2017 and 2018.

How Users Interact:

Clicking a bubble filters all other visuals to that category. The matrix drill-down arrow expands any category to product level. Right-clicking a category on Page 1 and selecting Drillthrough opens Page 2 with that category pre-filtered. The Reset button clears all active filters.

3. Performance Monitoring and Geographic Analysis



Design Choices

The decomposition tree analyses Total Revenue by customer_state, Category_English and DeliveryStatus.

The Avg Delivery Days KPI visual shows 9.35 days against a 10 day target with a YearMonth trend axis.

The On-Time Delivery Rate gauge shows 0.94 against a target of 0.90 on a scale of 0 to 1.

The YTD vs Previous Year area chart plots both YTD Revenue and Prev Year Revenue on the same axis using DimDate[YearMonth].

The map uses customer_state with bubble size encoding Total Orders and colour encoding Avg Review Score.

The key insight text box summarises the three main findings from this page for stakeholders who want a quick summary.

How Users Interact:

Clicking the plus icon on any decomposition tree node drills into the next dimension. Clicking a state bubble filters all other visuals to that state. The Reset button clears all active filters.

7. Insights and Recommendations

7.1 Analytical Insights

Revenue concentration risk

The top 3 product categories which are health beauty, watches gifts and bed bath table account for over 35% of total revenue. This means the business is heavily dependent on a small number of categories and any disruption to them would significantly impact overall performance.

Delivery speed drives customer satisfaction

The scatter chart on Page 2 shows a clear negative relationship between delivery days and review scores. Orders delivered in under 7 days average a review score above 4.2 while orders taking over 20 days average below 3.0. Faster delivery directly produces better customer satisfaction.

Geographic demand is heavily concentrated

São Paulo accounts for over 40% of all orders. The northern and northeastern states show very low order volumes despite large populations. This points to both a logistics challenge and an untapped market opportunity.

Credit card instalments dominate payments

Over 73% of orders are paid by credit card with a large proportion using multiple installment plans. While this drives higher average order values it also concentrates cash flow risk in the credit infrastructure.

Revenue growth slowed in mid-2018

The YoY Growth chart on Page 3 shows strong growth through 2017 and early 2018 followed by a visible slowdown from Q2 2018. The decomposition tree shows this slowdown is concentrated in the electronics category rather than being platform wide.

7.2 Recommendations

1. Olist should invest in last mile logistics partnerships in the northern and northeastern states. The scatter chart shows delivery time is the primary driver of

review scores. Reducing average delivery time by even 3 days in states like AM, PA and MA would meaningfully improve satisfaction scores and grow order volumes in currently underserved regions.

2. Olist should run a seller recruitment campaign targeting underrepresented categories such as electronics and auto accessories. Offering reduced commissions for the first 12 months would incentivise new sellers to join without requiring upfront marketing spend. This would reduce the revenue concentration risk identified in Insight 1.

3. Management should investigate the 2018 electronics slowdown as a priority. The decomposition tree analysis points to electronics as the main driver of the deceleration. A root cause analysis examining pricing competitiveness and emerging competitors should be conducted before the trend spreads to other categories.

8. Conclusion

The analysis of the Olist dataset reveals that Olist is a growing platform with strong overall performance. Total revenue of R$13.59M across 96,000 orders with an average review score of 4.09 out of 5 and an on-time delivery rate of 94% are all positive indicators.

However, the business faces three clear challenges. Revenue is too concentrated in a small number of categories, geographic demand is heavily skewed towards São Paulo leaving the north and northeast underserved, and the 2018 revenue slowdown in electronics needs urgent attention before it spreads.

The recommendations in this report point to logistics investment in underserved regions and category diversification as the two highest priority actions. Both are supported directly by the data and both have a clear measurable expected outcome based on the relationships identified in the dashboard.

Github Repository Link: https://github.com/FMuthoniMwangi/Brazilian_E-commerce_Analysis